1.27 Polar Coordinates: Introduction

Worked Solutions

1 a)

$A(2, \frac{\pi}{6})$
$B(4, \frac{5\pi}{12})$
$C(8, \frac{3\pi}{4})$
$D(3, -\frac{2\pi}{3})$
$E(5, -\frac{\pi}{2})$
$F(\frac{5}{2}, -\frac{\pi}{6})$

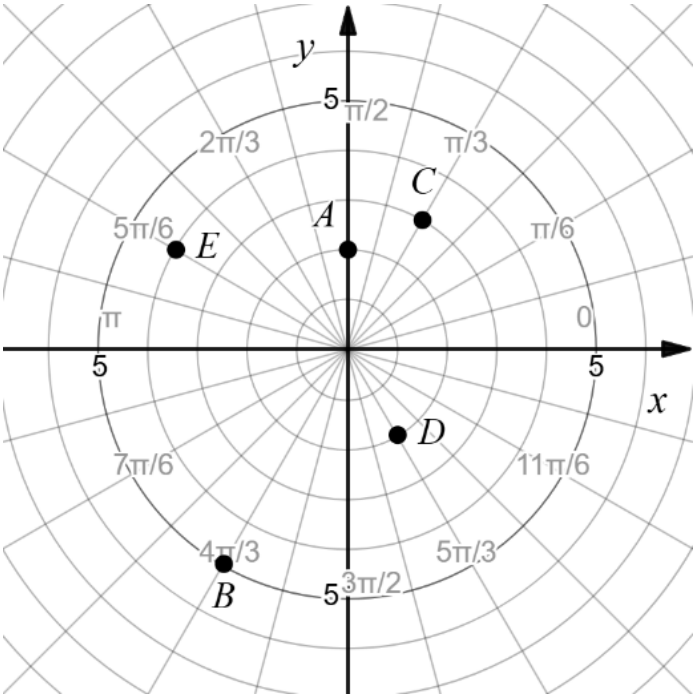
1 b)

C

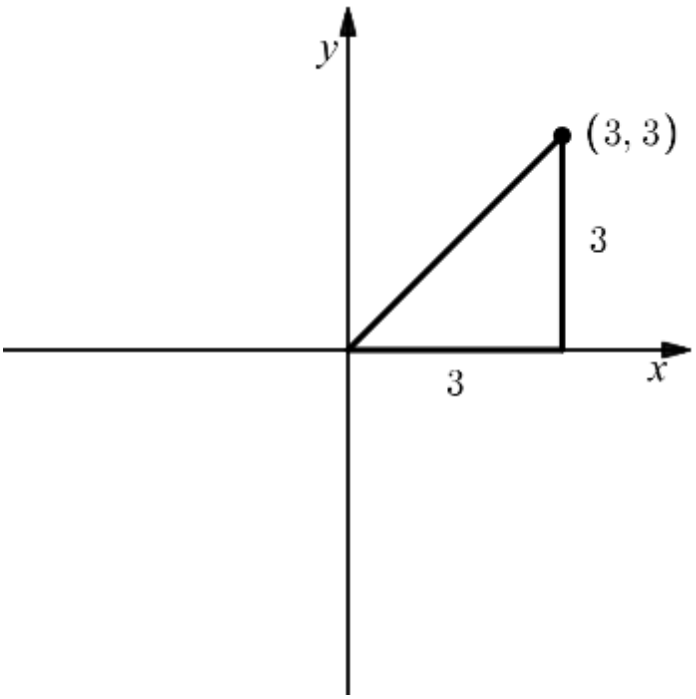
1 c)

A

2)



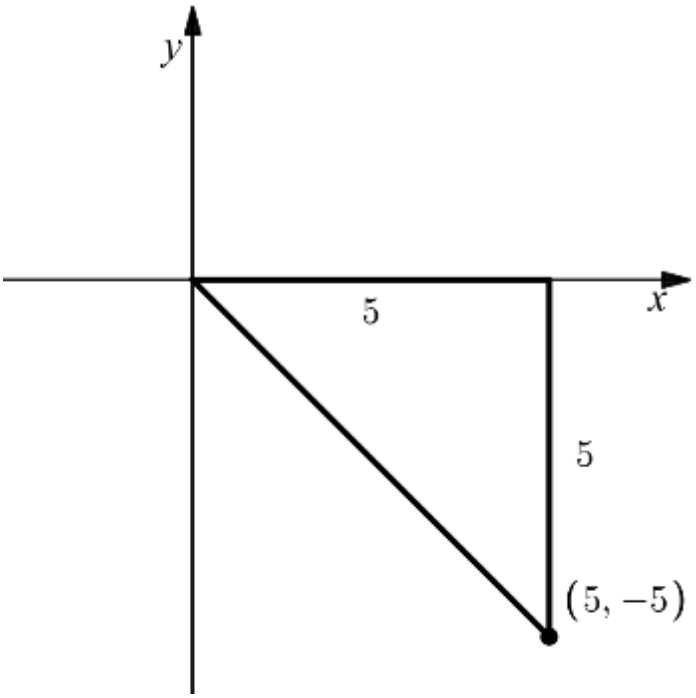
3 a)



$$\theta = \arctan(\tfrac{3}{3}) = \frac{\pi}{4}$$
$$r = \sqrt{3^2 + 3^2} = \sqrt{18} = 3\sqrt{2}$$

$(3\sqrt{2}, \frac{\pi}{4})$

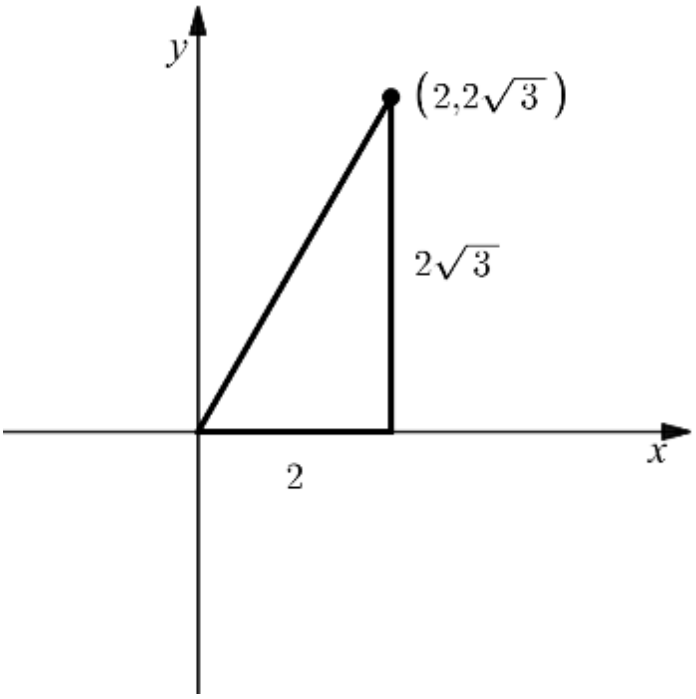
3 b)



$$\theta = \arctan(\tfrac{5}{5}) = \frac{\pi}{4}$$
$$r = \sqrt{5^2 + 5^2} = \sqrt{50} = 5\sqrt{2}$$

$(5\sqrt{2}, -\frac{\pi}{4})$

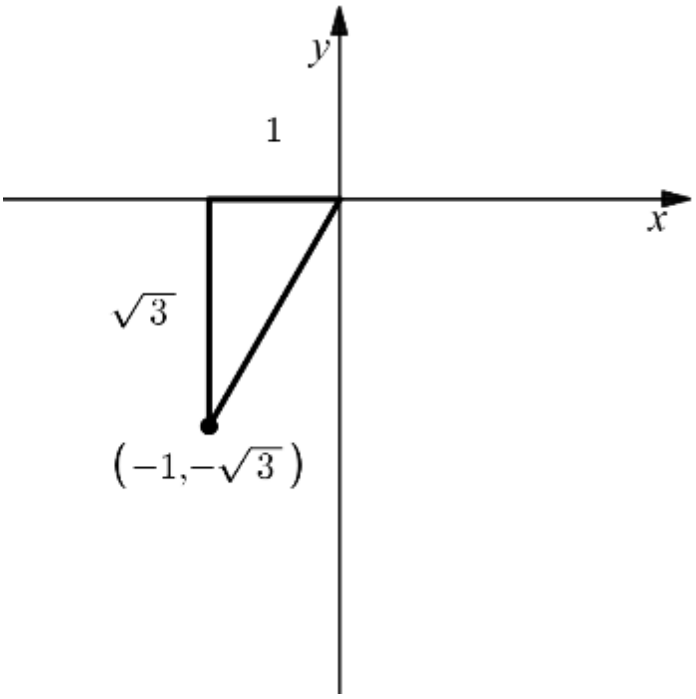
3 c)



$$\theta = \arctan(\tfrac{2\sqrt{3}}{2}) = \frac{\pi}{3}$$
$$r = \sqrt{2^2 + (2\sqrt{3})^2} = \sqrt{16} = 4$$

$(4, \frac{\pi}{3})$

3 d)

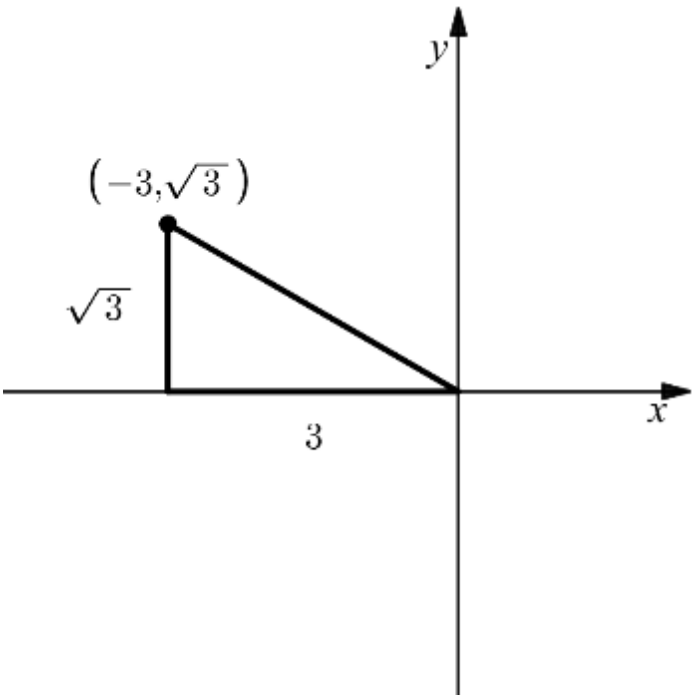


$$\theta = \arctan(\sqrt{3}) = \frac{\pi}{3}$$

$$r = \sqrt{1^2 + (\sqrt{3})^2} = \sqrt{4} = 2$$

$(2, -\frac{2\pi}{3})$

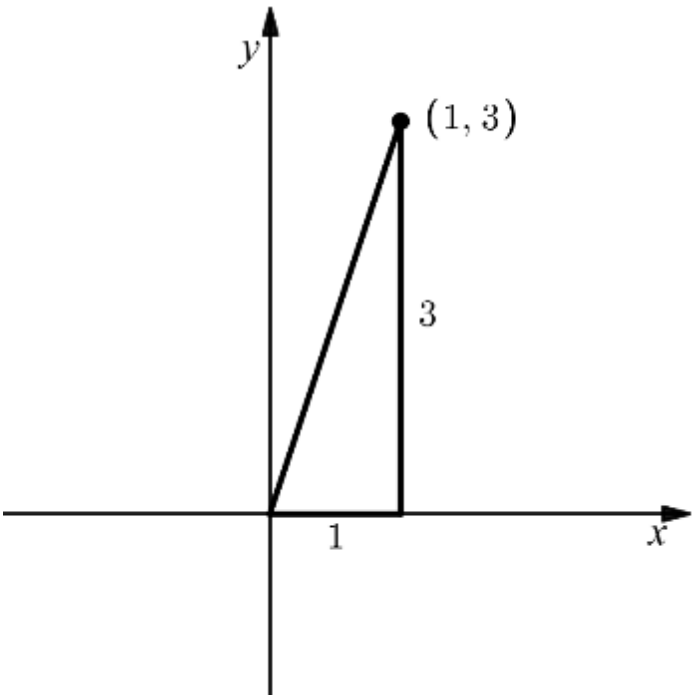
3 e)



$$\theta = \arctan(\tfrac{\sqrt{3}}{3}) = \frac{\pi}{6}$$
$$r = \sqrt{3^2 + (\sqrt{3})^2} = \sqrt{12} = 2\sqrt{3}$$

$(2\sqrt{3}, \frac{5\pi}{6})$

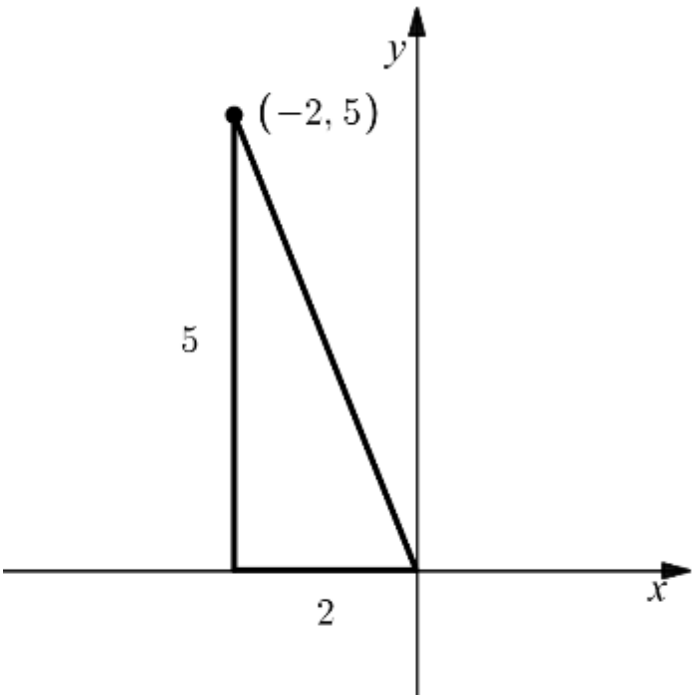
4 a)



$$\theta = \arctan(\tfrac{3}{1}) = 1.25$$
$$r = \sqrt{1^2 + 3^2} = \sqrt{10}$$

$(\sqrt{10}, 1.25)$

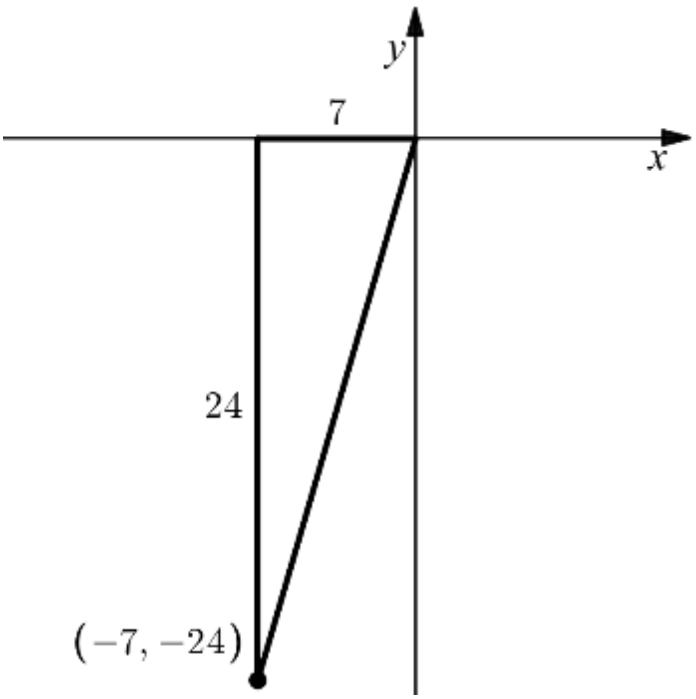
4 b)



$$\theta = \arctan(\tfrac{5}{2}) = 1.19$$
$$r = \sqrt{2^2 + 5^2} = \sqrt{29}$$

$(\sqrt{29}, 1.95)$

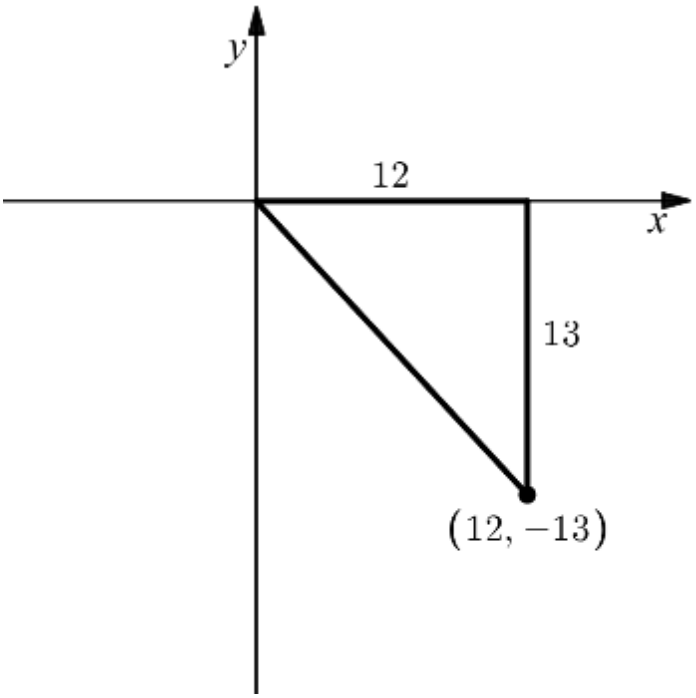
4 c)



$$\theta = \arctan(\tfrac{24}{7}) = 1.29$$
$$r = \sqrt{7^2 + 24^2} = \sqrt{225} = 25$$

$(25, -1.85)$

4 d)



$$\theta = \arctan(\tfrac{13}{12}) = 0.825$$
$$r = \sqrt{12^2 + 13^2} = \sqrt{313}$$

$(\sqrt{313}, -0.825)$

5 a)

$(2, -\frac{3\pi}{4})$

5 b)

$(1, -\frac{\pi}{6})$

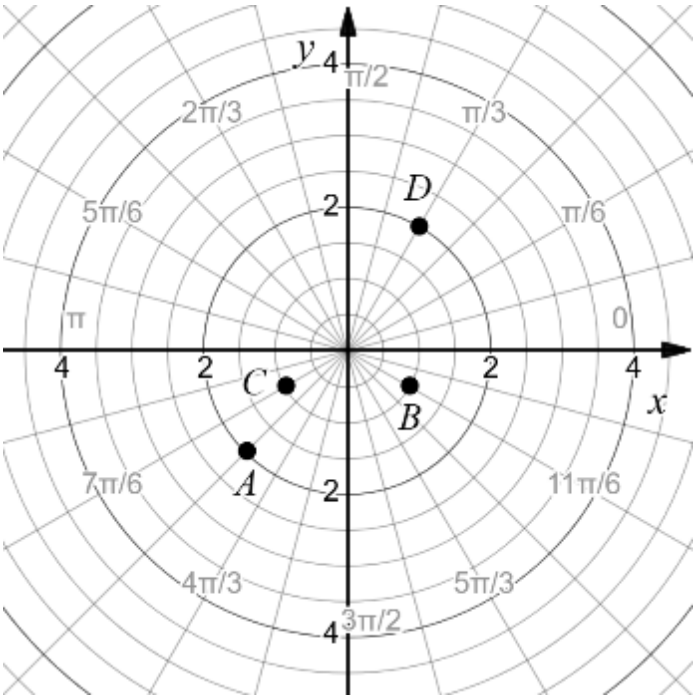
5 c)

$(1, -\frac{5\pi}{6})$

5 d)

$(2, \frac{\pi}{3})$

5 e)



6 a)

$(3\cos\frac{\pi}{4}, 3\sin\frac{\pi}{4}) = (\frac{3\sqrt{2}}{2}, \frac{3\sqrt{2}}{2})$

6 b)

$(2\cos\frac{5\pi}{6}, 2\sin\frac{5\pi}{6}) = (-\sqrt{3}, 1)$

6 c)

$(-4\cos\frac{\pi}{3}, -4\sin\frac{\pi}{3}) = (-2, -2\sqrt{3})$

7)

$(-3, \frac{\pi}{3} + \pi)$ so $p = \frac{4\pi}{3}$

$(3, \frac{\pi}{3} + 2\pi)$ so $q = \frac{7\pi}{3}$

8 a)

Each vertex is one rotation of $\frac{2\pi}{3}$ about the pole

$A(2, \frac{\pi}{6})$, $B(2, \frac{\pi}{6} + \frac{2\pi}{3})$, $C(2, \frac{\pi}{6} + \frac{4\pi}{3})$

So

$A(2, \frac{\pi}{6})$, $B(2, \frac{5\pi}{6})$, $C(2, -\frac{\pi}{2})$

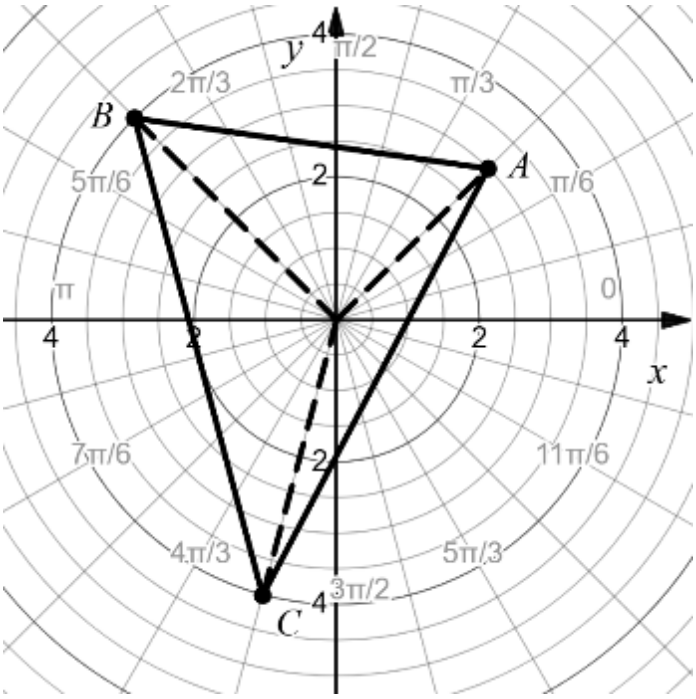
8 b)

$\frac{\pi}{3}$ is the angle to rotate from A to C about the pole

$$\frac{\pi}{6} + \frac{\pi}{3} = \frac{\pi}{2} \text{ or } \frac{\pi}{6} - \frac{\pi}{3} = -\frac{\pi}{6}$$

$A(2, \frac{\pi}{6})$, $B(0, 0)$, $C(2, \frac{\pi}{2})$ or $A(2, \frac{\pi}{6})$, $B(0, 0)$, $C(2, -\frac{\pi}{6})$

9)



$$\frac{1}{2}\times 3\times 4\times \sin(\frac{3\pi}{4}-\frac{\pi}{4})+\frac{1}{2}\times 4\times 4\times \sin(\frac{17\pi}{12}-\frac{3\pi}{4})$$
$$+\frac{1}{2}\times 4\times 3\times \sin(\frac{7\pi}{12}+\frac{\pi}{4})$$

$$= 6\sin\frac{\pi}{2}+8\sin\frac{2\pi}{3}+6\sin\frac{5\pi}{6}$$
$$= 6+4\sqrt{3}+3 = 9+4\sqrt{3} = 15.9 \text{ units}^2$$

10 a)

RHS: $4-3\sin\frac{\pi}{2}=4-3=1$ so the point lies on the line.

10 b)

Since $-1\le\sin\theta\le 1$

$$r = 4-3\times(-1) = 7$$

So when $\theta=\frac{3\pi}{2}$

$$(7,\tfrac{3\pi}{2})$$

10 c)

$r=4-3\sin\frac{\pi}{4}=4-\frac{3\sqrt{2}}{2}$

$$(4-\frac{3\sqrt{2}}{2})\cos\frac{\pi}{4}=(4-\frac{3\sqrt{2}}{2})\frac{\sqrt{2}}{2}=2\sqrt{2}-\frac{3}{2}$$

$$(4-\frac{3\sqrt{2}}{2})\sin\frac{\pi}{4}=2\sqrt{2}-\frac{3}{2}$$

So $(2\sqrt{2}-\frac{3}{2},2\sqrt{2}-\frac{3}{2})$

11)

The two angles must be related by:

$$\begin{aligned} b + \pi &= -b + 2\pi k \text{ for some integer } k \\ 2b &= 2\pi k - \pi \\ b &= \pi k - \tfrac{1}{2}\pi \\ \Rightarrow b &= \tfrac{1}{2}(2k-1)\pi \end{aligned}$$

That is, b must be some multiple of $(2n+1)\frac{\pi}{2}$ where $n \in \mathbb{Z}$

Or directly above or below the pole.